

AI for Financial Practitioners

Lecture 2 · Part I

From Models to Agentic Organization

How models become working systems through tools, skills, memory, agents and the deliberate organization of intelligent work

The central shift: intelligence acquires an organization

Lecture 1 organized the *cycle* of intelligent work. Lecture 2 organizes *who or what performs it*.



THE FINANCIAL ANALOGY

Banks already separate origination, approval, booking, settlement and audit. Asset managers separate research, portfolio construction, trading, risk and operations.

THE AI IMPLICATION

Agentic AI brings the same organizational question into software: divide responsibility according to evidence, expertise, permissions and control.

WHAT CHANGES

The design problem moves beyond “Which model?” toward decomposition, state, handoffs, escalation, verification and accountability.

EXECUTIVE QUESTION

How should intelligent work be organized so that capability becomes dependable institutional performance?

A model is a capability. A system gets the work completed.

A working system surrounds flexible intelligence with evidence, operations, state, controls and a route to completion.



WHY MODULARITY MATTERS

A stronger model can improve interpretation without changing database permissions, calculation engines, approval workflows or audit records. **WHY**

GOVERNANCE IMPROVES

When responsibilities are separated, failures become diagnosable: stale source, entity error, bad calculation, weak interpretation or failed review.

CREDIT COMMITTEE EXAMPLE

Default probability is only one component. The process also needs verified statements, exposure, collateral, covenants, prior decisions and delegated approval authority. **MANAGEMENT IMPLICATION**

Understand the work before selecting the model.

Model choice is one architectural decision, not the architecture itself.

Three extensions of intelligence: tools, skills and memory

They solve different problems and therefore require different controls.

TOOLS

Do

Extend the action space:
database query, pricing library,
calculator, code interpreter,
workflow interface.

CONTROL: permissions · parameters
· logging

SKILLS

Know how

Encode reusable professional
procedure: covenant review,
refinancing-risk review, due
diligence, valuation protocol.

CONTROL: versioning · tests ·
approved method

MEMORY

Continue

Preserve qualified state: last
filing reviewed, approved
assumptions, unresolved
exceptions, prior escalations.

CONTROL: provenance · scope ·
expiry · owner

Financial discipline: use deterministic tools for exact operations; encode stable professional methods as skills; store statements in memory according to their status—not merely because the model said them.

The agent: not a personality, a bounded unit of responsibility

An institutional agent receives an objective, observes state, chooses among permitted steps and must know when to stop or escalate.



WHERE AGENTS ADD VALUE

When the route cannot be specified efficiently in fixed rules: evidence differs by case, sources fail, documents conflict, or discoveries alter the next step. **WHERE THEY DO NOT**

If deterministic code performs the responsibility transparently, adding an agent may create discretion and coordination cost without adding information. **THE DESIGN QUESTION**

Where does bounded discretion create measurable value?

From one agent to an artificial organization

Multiple agents create value through meaningful specialization and independent information—not by multiplying artificial voices.



GOOD SPECIALIZATION

Different roles use distinct evidence, tools, methods or acceptance tests. A verifier recomputes; a citation checker opens the source; a red team searches for disconfirming evidence. **FALSE DIVERSITY**

Five agents reading the same erroneous summary do not create five independent opinions. Agreement can simply reproduce one upstream failure.

DISCIPLINED DISAGREEMENT

Preserve disagreement at the level of assumptions and claims—volume, price, costs, timing, management credibility—rather than averaging prose into artificial consensus. **HUMAN COMMITTEES, REDESIGNED**

The objective is not to simulate a meeting. It is to produce the evidence, challenge and escalation required by the responsibility.

The economics of decomposition: every specialist must earn its place

Divide work only where specialization or control exceeds coordination cost.

BENEFITS OF DECOMPOSITION

- ▶ distinct expertise and evidence
- ▶ parallel work and throughput
- ▶ independent verification
- ▶ clearer accountability and testing
- ▶ bounded permissions by role

COORDINATION COSTS

- ▶ handoffs and schema failures
- ▶ shared-state conflicts
- ▶ model/tool calls and latency
- ▶ integration and disagreement resolution
- ▶ additional human review burden

Marginal information or control

Marginal coordination cost

should justify each additional role

Default architecture: begin with one bounded agent + controlled tools. Add specialists only when their contribution is observable.

Worked design: an agentic investment memorandum

Separate evidence, financial analysis, market context, challenge, integration and human approval through explicit interfaces.



CASE STATE

Issuer graph, observation date and required output are protected common fields.

SPECIALIST OUTPUTS

Authorized filings; controlled debt/liquidity metrics; timestamped market observations; qualitative developments.

CHALLENGE

Checks entailment, contrary evidence, reconciliation and assumptions—without relying only on the polished draft.

INTEGRATION

Produces observations, interpretation, scenarios, uncertainty and analyst questions; unresolved challenges cannot disappear.

APPROVAL

The analyst accepts, modifies or rejects claims. Only qualified information is promoted to durable memory.

TEST

Role tests + interface failures + end-to-end scenarios + economic comparison with the current analyst process.

The responsibility map: the management artifact for agentic AI

Make responsibility visible before capability quietly becomes authority.

Responsibility	Evidence / input	Capability	May change	Acceptance / escalation	Owner
Evidence	Approved sources	Retrieval	Case evidence	Freshness / ambiguity	Research ops
Financials	Controlled data	Tools + skill	Calculated metrics	Reconcile / fail	Finance owner
Interpretation	Evidence metrics +	Agent	Draft claims	Cite / qualify	Analyst lead
Challenge	Sources claims +	Independent checks	Exceptions	Materiality gate	Risk / review
Integration	Specialist outputs	Agent	Draft memo	Preserve / dis-sent	Research lead
Approval	Full case	Human judgment	Approved state	Decision rights	Authorized analyst

WHAT THE MAP EXPOSES

Unassigned freshness checks, ambiguous memory ownership, exceptions with no destination, or agents whose separation has no business rationale.

WHY EXECUTIVES SHOULD CARE

It gives business, engineering, risk, compliance, finance and model-risk teams one common object for discussing the operating design.

From organized intelligence to accountable delegation

Agentic AI is a designed organization of intelligence. The next question is how much discretion that organization should receive.



1 · CAPABILITY

What can the system perceive, calculate, synthesize and do?

2 · RESPONSIBILITY

Which bounded unit owns each task, handoff, check and escalation?

3 · AUTHORITY

Which consequences may each role create without human approval?

Capability is not authority.

That distinction is the bridge to Lecture 2 · Part II: autonomy, governance and accountable delegation.